Power balance constraints:

$$\begin{aligned} P_k^{PV,used} + P_k^{PV,curt} &= \mathrm{P}_k^{PV} \\ P_k^{\downarrow} + P_k^{PV,used} &= P_k + \mathrm{P}_k^{rest} \end{aligned}$$

Electrical constraints:

$$\begin{aligned} 0 \leq P_k^{PV,used} &\leq \mathrm{P}_k^{PV} \\ 0 \leq P_k^{PV,curt} &\leq \mathrm{P}_k^{PV} \\ 0 \leq P_k^{\downarrow} &\leq \overline{\mathrm{B}} \end{aligned}$$

Comfort and Power constraints:

$$\underline{\mathrm{T}}^r \leq T_k^r \leq \overline{\mathrm{T}}^r$$

$$\underline{\mathrm{P}} \leq P_k \leq \overline{\mathrm{P}}$$

IV.2.10. MPC Operation

The optimization problem is solved at each control step using a receding horizon strategy. Only the first control action $T_0^{sp}, P_0^{PV,used}, P_0^{PV,curt}, P_0^{\downarrow}$ are actuated by the building management system. At the next control step, the state is updated with new measurements, and the optimization is solved again using updated forecasts.

IV.3. Pilot: EWH Cold Rooms

The pilot features four walk-in fridges and one walk-in freezer. Each of the four fridges contains a specific product type. There is one refrigeration cycle cooling the fridges, which is powered by two compressors: an on/off compressor and a modulated compressor. The refrigeration cycle of the freezer is powered by one on/off compressor. The flexibility of the system is enabled by computing dynamic setpoints for the 5 cold rooms decided by the electricity market, while keeping the temperatures strictly inside the food-safe ranges.

Although the fridges and freezer operate on independent cycles, they are modeled together for two primary reasons: Scalability: Integrating them into a single model ensures that any future hardware or operational changes can be adapted more efficiently. Control Logic: Since the flexibility manager reads

and actuates all assets simultaneously, it is more cohesive to bind them within a unified framework.

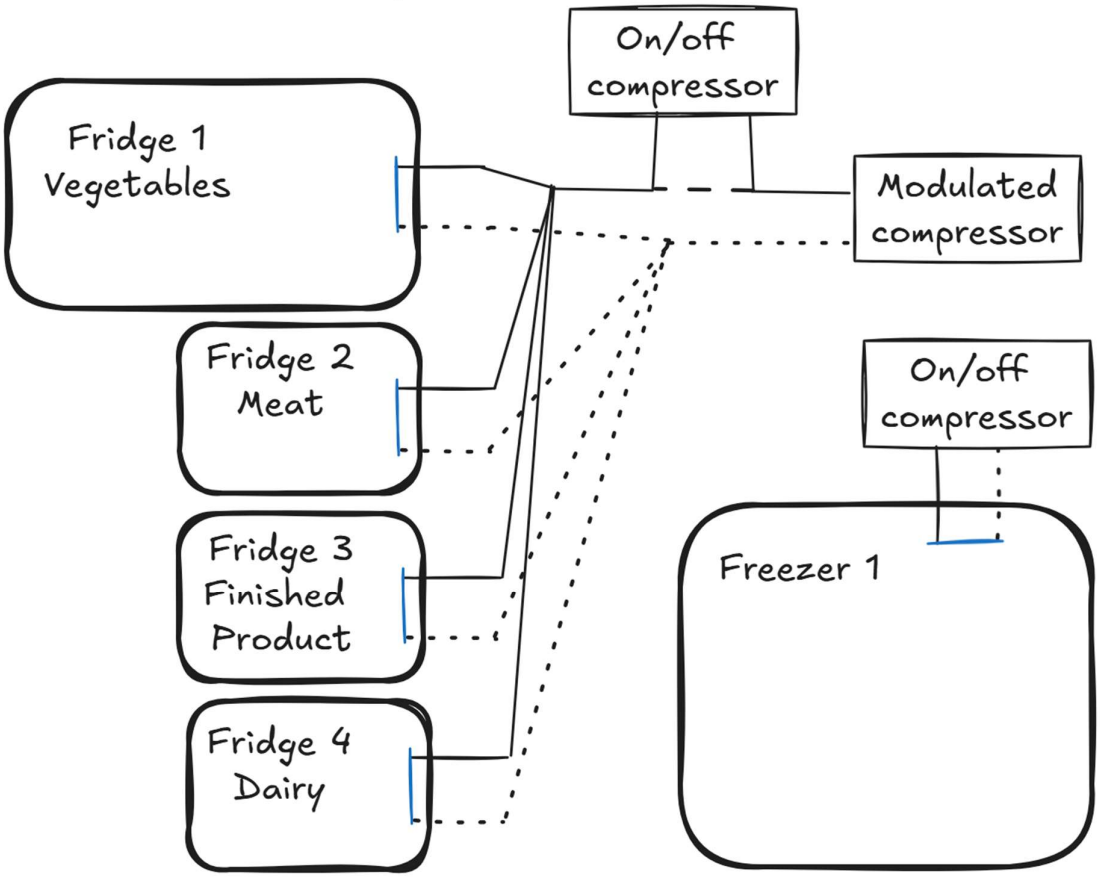

Figure 2. Simple diagram of the cold rooms and the related compressor setup. The dashed line indicates that the on/off compressor can be bypassed. The diagram does not reflect all parts of the refrigeration cycles.

IV.3.1. Sets and Indices

Symbol	Description
$\mathcal{R}$	Set of cold rooms
$r \in \mathcal{R}$	Cold room index
$k = 0, \dots, H_u - 1$	Control step
H_u	Number of time steps in the MPC horizon
Δt	Time step
$T = \Delta t \cdot H_u$	Control horizon time

IV.3.2. State Variables

Symbol	Description
T_k^r	Temperature of cold room r at control step k
T_k	Stacked vector of cold room temperatures

IV.3.3. Control Variables

Symbol	Description
$T_k^{r,sp}$	Temperature setpoint for cold room r
T_k^{sp}	Stacked vector of temperature setpoints
p_k^1	Electrical power consumption of the modulated fridge compressor at time k
δ_k^{fridge}	Binary variable indicating if the fridge on/off compressor is active at time k
$\delta_k^{freezer}$	Binary variable indicating if the freezer on/off compressor is active at time k
u_k^r	Allocated electrical power for cooling of cold room $\forall r \in \mathcal{R}\backslash\{5\}$ at time k
$u_k^{r,req}$	Requested electrical power for cooling of cold room $\forall r \in \mathcal{R}\backslash\{5\}$ at time k
$P_k^{\uparrow}$	Power sold to the grid
$P_k^{\downarrow}$	Power bought from the grid
$P_k^{PV,used}$	PV power used for self-consumption
$P_k^{PV,curt}$	PV power curtailed

IV.3.4. Exogenous State - Origin: Forecasting Service

Symbol	Description
T_k^a	Ambient temperature
$Q_k^{r,door}$	Energy added to cold room r at time k if the door is open
Q_k^{door}	Stacked door openings
P_k^{PV}	PV production forecast
P_k^{rest}	Power consumed by non-controlled assets (rest of hotel)

We define the stacked disturbances $\mathrm{d}_k = [\mathrm{T}_k^{soil},\ Q_k^{r^\intercal}]^\intercal$.

IV.3.5. Exogenous State - Origin: Trading Manager

Symbol	Description
$\pi_k^{\downarrow}$	Electricity purchase price (ToU)
$\pi_k^{\uparrow}$	Electricity sell price (ToU)

IV.3.6. Algebraic

Symbol	Description
$p_k^2 = \delta_k^{fridge} p_{max}^2$	Electrical power consumption of compressor (fridge on/off)
$p_k^3 = \delta_k^{freezer} p_{max}^3$	Electrical power consumption of compressor (freezer on/off)
$P_k = \sum_i p_k^i$	Total power used by the three compressors
$U_k^{req} = \sum_{r \in \mathcal{R}\backslash\{5\}} u_k^{r,req}$	Total requested electrical power for the cooling of the four fridges

IV.3.7. Parameters (Metadata)

Symbol	Description
$\overline{\mathrm{B}}$	Grid purchase limit
$\overline{\mathrm{S}}$	Grid selling limit
$\underline{\mathrm{T}}^r$	Cold room r temperature lower bound
$\overline{\mathrm{T}}^r$	Cold room r temperature upper bound
$\underline{\mathrm{p}}$	Power consumption lower limit
$\overline{\mathrm{p}}$	Power consumption upper limit
K^{gain}	The electrical power [kW] needed per °C of temperature deviation to drive the cold room from its current temperature toward the setpoint.
M^{fridge}	Large constant used to activate or deactivate the big-M constraint of the fridges
$M^{freezer}$	Large constant used to activate or deactivate the big-M constraint of the freezer

IV.3.8. Digital Twin

The evolution of the cold room state and resulting power consumption are governed by the Digital Twin constraints. The dynamics of the cold rooms are derived from the first law of thermodynamics applied to a lumped thermal mass. This results in the energy balance of a single room.

$$mc_v \frac{dT(t)}{dt} = Q^{ambient}(t) + Q^{door}(t) - Q^{cool}(t)$$

where m [kg] is the effective thermal mass of the room, and c_v $[\frac{\mathrm{J}}{\mathrm{kg\ K}}]$ is the specific heat capacity of the thermal mass. The first term approximates the heat transfer with the surrounding through walls, ceiling, and floor (the envelope of the room), given as

$$Q^{ambient}(t) = UA(T^a(t) - T(t))$$

where U $[\frac{\mathrm{W}}{\mathrm{m^2K}}]$ is the overall transfer coefficient of the rooms and A $[\mathrm{m}^2]$ is the effective surface area of the room envelope.

The second term is the energy added to the room while the door is open. This will be a disturbance and estimated empirically.

The third term is the energy removed by the refrigeration cycle. The energy removed is calculated using a compressor coefficient of performance (COP $[-]$) as

$$Q^{cool} = COP \cdot P(t)$$

where $P(t)$ [kW] is the power consumed by a compressor.

State space form

The proposed model for all five cold rooms can be written in continuous time on state space form as

$$\dot{x}(t) = Ax(t) + Bu(t) + Ed(t)$$

Where

$$x(t) = [T^1(t) \quad T^2(t) \quad T^3(t) \quad T^4(t) \quad T^5(t)]^T$$
$$u(t) = [u^1(t) \quad u^2(t) \quad u^3(t) \quad u^4(t) \quad u^5(t)]^T$$
$$d(t) = [T^a(t) \quad Q^{1,door}(t) \quad Q^{2,door}(t) \quad Q^{3,door}(t) \quad Q^{4,door}(t) \quad Q^{5,door}(t)]^T$$

and the matrices are defined as

$$A = \mathrm{diag}\left(-\frac{U^1A^1}{m^1c_v^1}, \dots, -\frac{U^rA^r}{m^rc_v^r}\right)$$
$$B = \mathrm{diag}\left(-\frac{COP^1}{m^1c_v^1}, \dots, -\frac{COP^r}{m^rc_v^r}\right)$$
$$E = \begin{bmatrix} \frac{U^1A^1}{m^1c_v^1} & \frac{1}{m^1c_v^1} & 0 & \cdots & 0 \\ \frac{U^2A^2}{m^2c_v^2} & 0 & \frac{1}{m^2c_v^2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ \frac{U^rA^r}{m^rc_v^r} & 0 & \cdots & 0 & \frac{1}{m^rc_v^r} \end{bmatrix}$$

Discrete time state space model

To use the digital twin in the flexibility optimization it should be discretized. Applying a zero-order hold for the control and disturbance variables, the system can be discretized with a sampling time Δt giving

$$x_{k+1} = A_d x_k + B_d u_k + E_d d_k$$

where

$$A_d = e^{AT_s}$$
$$B_d = \int_0^{T_s} e^{A\tau} B d\tau$$
$$E_d = \int_0^{T_s} e^{A\tau} E d\tau$$

The easy calculation of these is done as

$$\begin{bmatrix} A_d & B_d & E_d \\ 0 & I & 0 \\ 0 & 0 & I \end{bmatrix} = \exp\left(\begin{bmatrix} A & B & E \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} T_s\right)$$

Replication of the compressor controller

As we will only be able to control the setpoints of the cold rooms, a model of the compressor logic needs to be developed for the flexibility optimization. The true physical system driving the compressor power usage is complex and include valve openings and pressures in the refrigeration cycle. To get a suitable model for the optimization we abstract these low-level dynamics by assuming a hierarchical control structure of the compressors driven by the differences in the room temperatures and setpoints.

The on/off compressor of the freezer is switched on and off given a hysteresis band defined as

$$\text{Compressor ON} \quad \text{if} \quad T_k^5 \geq T_k^{5,sp} + d$$
$$\text{Compressor OFF} \quad \text{if} \quad T_k^5 \leq T_k^{5,sp} - d$$

where d $[^\circ C]$ is the bandwidth. This can be written as constraints using a big-M notation as

$$T_k - (T_{set,k} + d) \leq M^{freezer} \delta_k^{freezer}$$
$$(T_{set,k} - d) - T_k \leq M^{freezer}(1 - \delta_k^{freezer})$$

giving

$$u_k^5 = \delta_k^{freezer} P_{max}^3$$

The logic for the fridge system is more advanced, as the two compressors in that refrigeration cycle supply cooling for all four fridges. This is handled by calculating the requested power needed for cooling by each of the four fridges and then letting the optimizer delegate the available power for cooling. The requested power is not a variable related to the true system. It is an abstraction that allows the model to handle the distribution of cooling in the fridges without having the knowledge of the expansion valves in the true refrigeration cycle.

The requested electrical power for cooling each of the fridges is given as

$$u_k^{r,req} \geq K^{gain}(T_k^r - T_k^{r,sp})$$

where K^{gain} is a parameter translating the desired temperature change to an electrical power request for the compressors.

The primary modulated compressor ramps up first to meet demand. If the load exceeds its limit, the second on/off compressor kicks in to provide additional power. To model this behavior the primary modulated compressor is a control variable p_k^1 with the constraints

$$0 \leq p_k^1 \leq P_{max}^1$$
$$p_k^1 \leq U_k^{req}$$

The logic for activating the on/off compressor for the fridges the following big-M constraint is used

$$U_k^{req} - P_{max}^1 \leq M^{fridge} \delta_k^{fridge}$$
$$p_k^2 = \delta_k^{fridge} P_{max}^2$$

The allocation to each of the four rooms is then constrained by

$$\sum_{r \in \mathcal{R} \setminus \{5\}} u_k^r = p_k^1 + p_k^2$$
$$0 \leq u_k^r \leq u_k^{r,req}, \quad \forall r \in \mathcal{R} \setminus \{5\}$$

Simulation code for the cold-room thermal dynamics and compressor-control logic is available in the BlueBird GitHub repository at the path: flexibility-manager/dt/EWH_coldrooms.

IV.3.9. Model Predictive Control

The receding horizon optimization problem reads as:

$$\underset{\substack{T^{sp},u^{r},u^{r,req},\delta^{freezer},\delta^{fridge}\\ P^{PV,used},P^{PV,curt}\\ P^{\uparrow},P^{\downarrow}}}{\text{minimize}} \sum_{k=0}^{H_u-1} \left(P_k^{\downarrow}\pi_k^{\downarrow} - P_k^{\uparrow}\pi_k^{\uparrow}\right)\Delta t$$

Subject to ($\forall k = 0, \dots, H_u - 1$):

Power Balance Constraints:

$$P_k^{PV,used} + P_k^{\uparrow} + P_k^{PV,curt} = \mathrm{P}_k^{PV}$$
$$P_k^{\downarrow} + P_k^{PV,used} = P_k + \mathrm{P}_k^{rest}$$

Electrical Constraints:

$$0 \le P_k^{PV,used} \le P_k^{PV}$$
$$0 \le P_k^{PV,curt} \le P_k^{PV}$$
$$0 \le P_k^{\uparrow} \le \overline{\mathrm{S}}$$
$$0 \le P_k^{\downarrow} \le \overline{\mathrm{B}}$$
$$P_k^{\uparrow} \cdot P_k^{\downarrow} = 0$$

State space Constraints:

$$x_{k+1} = A_d x_k + B_d u_k + E_d d_k$$
$$x_0 = x^{inti}$$

Compressor logic Constraints :

$$u_k^{r,req} \ge K^{gain}(T_k^r - T_k^{r,sp})$$
$$T_k - (T_{set,k} + d) \le M^{freezer}\delta_k^{freezer}$$
$$(T_{set,k} - d) - T_k \le M^{freezer}(1 - \delta_k^{freezer})$$
$$U_k^{req} - \overline{\mathrm{P}}^1 \le M^{fridge}\delta_k^{fridge}$$
$$\delta^{freezer} \in \{0,1\}$$
$$\delta^{fridge} \in \{0,1\}$$

Algebraic:

$$u_k^5 = p_k^3 = \delta_k^{freezer}\overline{\mathrm{P}}^3$$
$$p_k^2 = \delta_k^{fridge}\overline{\mathrm{P}}^2$$
$$\sum_{r\in\mathcal{R}\setminus\{5\}} u_k^r = p_k^1 + p_k^2$$
$$P_k = p_k^1 + p_k^2 + p_k^3$$

Bounds:

$$p_k^1 \le U_k^{req}$$
$$\underline{\mathrm{P}}^1 \le p_k^1 \le \overline{\mathrm{P}}^1$$
$$0 \le u_k^r \le u_k^{r,req}, \quad \forall r \in \mathcal{R}\setminus\{5\}$$

Comfort Constraints:

$$\underline{\mathrm{T}}^r \le x_k^r \le \overline{\mathrm{T}}^r$$